Supplementary Information for: Transition Certification with Composite Indices: The Acceptance Gap Between Compensatory and Noncompensatory Rules

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Abstract

This Supplementary Information provides additional proofs, the CES diagonal thresholds and pointwise off-diagonal criterion, the specified disturbance variants, the synthetic biomass illustration, and practitioner guidance. It also states the scope of the finite numerical checks and formalization. The Northern-cod values are used only for a historical dimensional comparison; no parameter is estimated, and no fit, forecast, validation, or empirical performance claim is made.

S1. Abbreviations

Table S1: Abbreviations.

CES	constant elasticity of substitution
DFO	Fisheries and Oceans Canada
LPI	Living Planet Index
LRP	limit reference point
MCDA	multi-criteria decision analysis
MSY	maximum sustainable yield
NCAM	Northern Cod Assessment Model
SI	Supplementary Information

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S2. Remaining proofs

S2.1. Persistence under hold-prefix extension (main-text Proposition 3(i))

Proof. (Hierarchy.) Backward induction on stages. Base: at the final stage the terminal sets satisfy $G \subseteq G^w \subseteq G^{\text{phys}}$ by main-text Lemma 2 and the definition of the terminal sets. Step: the accepted set of a stage is the set of states from which some action keeps the tube in the safe set and the successor in the next-stage accepted set; applying main-text Proposition 1(i) to the stage's action sets and the induction hypothesis to the next-stage accepted sets preserves the three-way inclusion at the stage level. (Persistence of strictness.) With HOLD the unique prefix action, a prefix state is accepted exactly when it lies in the prefix safe set and in the next-stage accepted set; the stage-0 regions are therefore the witness regions of main-text Theorem 2 pulled back through the (identity) holds, and the strictness witnesses ($I \neq \varnothing$ and the point $(\frac{1}{2}, \frac{1}{10}, \frac{1}{10})$) persist. (Conditionality.) The example below erases the gap because the repair stage's accepted sets cover every first-stage successor while the first-stage tubes remain safe and the original final terminal rules are preserved. A safe continuation cannot repair an earlier tube breach; coincident terminal sets alone do not align path constraints; and coupling actions across stages alone is not sufficient. No general multi-stage erasure rule is asserted. □

S2.2. The erasure datum (main-text Proposition 3(ii))

Shared terminal rules. There are two margins and no reserve. At the end of the final interval the typed terminal set is $G = \{s_1 \geq 0, s_2 \geq 0\}$, while the scalarized terminal set at weight w is $G^w = \{w \cdot s \geq 0\}$. These are the same rules in the one-stage truncation and the two-stage extension; they are not declared coincident.

Stage 2, the repair interval. Its path-safe set is all of $\mathbb{R}^2$. From every starting state s, the single action REPAIR has tube $\{s\}$ and successor $(1, 1)$, which lies in both G and every G^w for $w \in W_+$. Consequently the stage-2 typed accepted set and every stage-2 scalarized accepted set are all of $\mathbb{R}^2$.

Stage 1. Its path-safe set is the nonnegative orthant, and the two actions have constant tubes at the initial state, with successors

$$A_1(s) = (s_1 - 1, s_2 + 1), \qquad A_2(s) = (s_1 + 1, s_2 - 1).$$

At $z^* = (2/5, 2/5)$ both stage-1 tubes are safe. In the two-stage problem, both successors lie in the stage-2 accepted sets $\mathbb{R}^2$, so each first-stage action is accepted under both doctrines. Backward induction therefore accepts z^*

as typed and weak in the extension. No earlier path breach is repaired; the first-stage path is safe and the repair stage enlarges only the continuation accepted sets.

One-stage truncation with the same terminal rules. Removing the repair interval leaves G and G^w as the terminal tests after stage 1. Both successors fail the typed terminal test, since their negative coordinate is $-3/5$. For $r = w_2/w_1$ with $w_1, w_2 > 0$, A_1 is scalarized-admissible exactly when $-3/5 + (7/5)r \geq 0$, i.e. $r \geq 3/7$; A_2 is admissible exactly when $7/5 - (3/5)r \geq 0$, i.e. $r \leq 7/3$. These intervals cover all positive ratios, and the coordinate weights are covered directly by A_2 at $r = 0$ and A_1 at $r = +\infty$. Thus z^* is weak-accepted but typed-rejected in the truncation. The difference is precisely the removal of a repair stage whose accepted sets covered the stage-1 successors, not a change in terminal semantics.

S2.3. The substitutability ladder (main-text Theorem 4)

Notation as in main-text Section 4.5: $\lambda_i = 1 + s_i$, normalised weights w, $M_\theta(\lambda; w) = (\sum_i w_i \lambda_i^\theta)^{1/\theta}$, $\sigma = 1/(1-\theta)$. The collapse convention is an additional floor-safety rule, not a standard CES identity.

Lemma S1 (the handshake)**.** *With weights normalised, $\sum_i w_i \lambda_i \geq 1 \iff w \cdot s \geq 0$; the $\theta = 1$ member therefore reproduces the compensatory operator of the main text exactly – the same acceptance set at every state and weight, and at the witness point the same per-weight plan assignment ($\rho_1 = 2/3$, $\rho_2 = 3/2$).*

Proof. $\sum_i w_i \lambda_i = \sum_i w_i + \sum_i w_i s_i = 1 + w \cdot s$. □

Lemma S2 (collapse convention)**.** *For every $\theta < 1$ member, any tube value $\lambda_i \leq 0$ rejects the plan: a coordinate at or below the collapse level cannot be compensated by surpluses elsewhere.*

This rejection rule is imposed uniformly, not derived from every power mean. At $\theta = 0$, a zero input with positive weight makes the geometric mean zero; at $\theta < 0$, zero is outside the usual power-mean domain. For $0 < \theta < 1$, however, the mean can remain positive when one coordinate is zero (for example, equal weights on $(0, 4)$ give $M_{1/2} = 1$), so automatic rejection is an additional safety convention; negative inputs are not standard geometric-mean inputs. The $\theta = 1$ member alone is allowed to compensate across collapse: at $(1/2, 1, 1)$ the fast trough has $\lambda_1 = 0$ and is rejected at every $\theta < 1$ by convention, while the linear family retains its declared compensation rule.

The master equation. On the diagonal $z = (x < 1, s, s)$ with $s \in (1, 2)$, FAST's worst-case trough is $\lambda = (s-1, s+1)$ and SLOW's is $(s+1, s-1)$; STAGED is inadmissible ($x < 1$). Write $p = w_1$ for the weight share, $A = (s-1)^\theta$, $B = (s+1)^\theta$. For $\theta > 0$, FAST is accepted at p iff $pA + (1-p)B \geq 1$ and SLOW iff $pB + (1-p)A \geq 1$; both are affine in p and sum to $A+B$. FAST's acceptance set is a downward interval $[0, p_F]$ (its condition is decreasing in p since $A < B$) and SLOW's an upward interval $[p_S, 1]$, with $p_S = 1 - p_F$ by symmetry; the cover $[0, p_F] \cup [p_S, 1] = [0, 1]$ holds iff $p_F \geq \frac{1}{2}$, i.e. iff the condition holds at $p = \frac{1}{2}$: $(A + B)/2 \geq 1$. For $\theta < 0$ the inequality $M_\theta \geq 1$ reads $\sum w_i \lambda_i^\theta \leq 1$ and the same argument gives $(A + B)/2 \leq 1$. For $\theta = 0$ the condition is $(s-1)^p (s+1)^{1-p} \geq 1$ with the cover at $p = \frac{1}{2}$: $s^2 - 1 \geq 1$. This is the averaging argument: the two plan conditions sum to a weight-independent quantity, so the equal weight is always the binding one.

Proof of main-text Theorem 4. (i) *Nesting.* For fixed positive λ and weights, M_θ is nondecreasing in θ (the power-mean inequality); the collapse convention preserves this, since acceptance at any $\theta' < 1$ forces $\lambda > 0$ on the serving plan's tube, where the family is defined. Per plan and weight, hence per weight, hence for the protocol: acceptance at θ' implies acceptance at every $\theta > \theta'$, so the accepted rungs form a prefix and $\sigma^*(z)$ is well defined. *Finiteness* on gap states with both coordinates in $(1, 2)$: at the equal weight, FAST's condition contains the term $(s_1 - 1)^\theta$ and SLOW's the term $(s_2 - 1)^\theta$, both of which tend to ∞ as $\theta \to -\infty$ because $s_i - 1 < 1$; so for some finite θ neither plan serves the equal weight and the state is rejected (on the diagonal this is $(s-1)^\theta \to \infty$). *Positivity*: $s_1 + s_2 \geq 2$ accepts at $\theta = 1$. *Only-linear states*: if $\min(s_1, s_2) \leq 1$, say $s_1 \leq 1$, then FAST's trough has $\lambda_1 = s_1 - 1 \leq 0$ and is rejected at every $\theta < 1$ by Lemma S2; SLOW alone must then cover every weight, including the weight concentrated on coordinate 2, where its condition is $s_2 - 1 \geq 1$, i.e. $s_2 \geq 2$, excluded on a gap state. So only the linear member certifies and $\sigma^* = \infty$. (ii) Immediate from (i): every gap state is linear-accepted, the accepted rungs form a prefix, and finiteness (or only-linearity) supplies a positive elasticity at which the state is rejected. On the deep-interior subset ($1 < s_1 < 2$, $1 < s_2 < 2$, $s_1 + s_2 > 2$), write $p = w_1 \in [0, 1]$ and define the linear slacks $L_F(p) = p(s_1 - 2) + (1-p)s_2$ and $L_S(p) = ps_1 + (1-p)(s_2 - 2)$. The licensing intervals overlap strictly because $s_1 + s_2 > 2$, so $\Delta(p) = \max\{L_F(p), L_S(p)\} > 0$ for every p. The function Δ is continuous on the compact simplex $[0, 1]$, hence $\eta = \min_{p \in [0,1]} \Delta(p) > 0$. All tube inputs $1 + s_i - d_{a,i}$ are positive on this deep-interior set. The best-plan CES margin is jointly continuous in (θ, p) near $\theta = 1$ on this compact

positive domain; uniform continuity therefore gives an $\varepsilon > 0$ such that the cover remains strict for some $\theta \in (1-\varepsilon, 1)$. This supplies a finite positive elasticity $\sigma = 1/(1-\theta)$ below the linear limit that still accepts the state. On gap states with $\min(s_1, s_2) \le 1$, the collapse convention rejects every $\theta < 1$ member, so only the linear member accepts. (iii) Evaluate the master comparison at $s = 2$: for $\theta \in (0, 1)$ the accepting side is ≥ 2 and $(s-1)^\theta + (s+1)^\theta = 1 + 3^\theta > 2$ strictly; for $\theta < 0$ the accepting side is ≤ 2 and $1 + 3^\theta < 2$ strictly; for $\theta = 0$, $4 > 2$. All strictly accepting at $s = 2$, so by continuity some $\varepsilon > 0$ has every $s \in (2-\varepsilon, 2)$ accepted at that rung, and these are gap states (typed acceptance needs $s \ge 2$); the acceptance condition is monotone in s on $(1, 2)$ (both A and B move the accepting way), so the accepted set is exactly $[s^*(\theta), 2)$ with $s^*(\theta) < 2$. The closed forms of the displayed rungs in main-text Table 7 follow by direct evaluation ($\theta = 2/3$: $s^* = \sqrt{6\sqrt{3} - 9}$; $\theta = 1/2$: $\sqrt{s-1} + \sqrt{s+1} \ge 2$, equality at $s = 5/4$; $\theta = 0$: $s^2 \ge 2$; $\theta = -1$: $(s^2 - 1)/s \ge 1$; $\theta = -2$: $(s+1)^2 + (s-1)^2 \le 2(s^2-1)^2 \iff s^2 \ge 3$; $\theta = -m$: $(s-1)^{-m} + (s+1)^{-m} \le 2$). Nesting gives nondecreasing critical floors as elasticity falls; the strict inequalities among the displayed rungs follow from these explicit roots and brackets, not from nesting alone. At $\theta \to -\infty$, the power mean tends to the minimum over the positive-weight support. A strictly positive full-support weight therefore requires every coordinate to meet its floor along the tube, while typed safety passes every support minimum; the Leontief-limit all-weight accepted set is exactly the typed set. Hence $\inf_z \sigma^*(z) = 0$ and only $\sigma = 0$ is uniformly safe. □

S2.4. The common-shock verdict (main-text Proposition 6)

The variant is defined in main-text Section 4.6 and SI S4 below. Write $A(p) = ps_1 + (1-p)s_2$ for the aggregate margin at weight share p.

Proof sketch. (i) FAST's common-class requirement is $A(p) \ge \max(2p,\ p + \frac{1}{2})$ (its own tent $3/2$ in s_1 plus the s_1-event, and the s_2-event weighted by $1-p$); SLOW's is the mirror; STAGED's ($x \ge 1$) is $A(p) \ge \max(p, 1-p)/2 - 1/8$, the event exposure of a growing floor (kinked in p; its admissibility decomposes exactly into the two affine conditions of the kink's branches). The pointwise minimum of the two principal requirements is a tent with its only kink at $p = \frac{1}{2}$, so an affine A covers it exactly at the three abscissae $p \in \{0, \frac{1}{2}, 1\}$; the closed forms follow: typed acceptance needs $s_1 \ge 2$ with $s_2 \ge \frac{1}{2}$ (or the mirror) or the staged branch $x \ge 1$, $\min(s) \ge \frac{3}{8}$; weak acceptance with $x < 1$ needs $s_1 \ge \frac{1}{2}$, $s_2 \ge \frac{1}{2}$ (the endpoints $p = 0, 1$) and $s_1 + s_2 \ge 2$ (the kink). The licensing endpoint of FAST on the surviving region equals its action-indexed endpoint $1/(1+\rho_1)$, since the binding branch there is the own-tent branch.

The closed forms above follow from the stated pointwise maxima and the three critical abscissae. (ii) The augmented menu adds STAGED_κ, admissible at $x + \kappa \geq 1$ with the same floor requirement $s_i \geq \frac{3}{8}$; κ cannot move the floors, so on $\{\min(s) < \frac{3}{8}\} \setminus \mathcal{V}^{\text{cs}}_{\text{typ}}$ no finite κ certifies. On the complement the requirement is the action-indexed one, $\kappa = (1 - x)_+$. The 3/8 boundary is structural: the event exposure of a growing floor is $\max(0, \delta - 1/8) = 3/8$ at $\delta = 1/2$. (iii) The blend's s_1-trough is $s_1 - \delta - \delta_{\text{blend}} \cdot \frac{3}{2}$ and its s_2-trough $s_2 - \delta - (1 - \delta_{\text{blend}}) \cdot \frac{3}{2}$ (each coordinate takes its event plus the corresponding fraction of the principal tent); the window is the nonnegativity interval of the two intersected with $[0, 1]$, nonempty iff $s_1 \geq \delta$, $s_2 \geq \delta$ and $s_1 + s_2 \geq \frac{3}{2} + 2\delta$. □

S3. The substitutability spectrum in full

This section records the CES results summarized in main-text Section 4.5 and Theorem 4. Results concern the witness datum and the explicit collapse convention; σ denotes the elasticity of substitution, while the Schaefer surplus in SI S5 is written ς.

S3.1. The critical-elasticity landscape on the diagonal

For each gap state the accepted rungs form a prefix of the ladder (main-text Theorem 4(i)), so the σ-aggregator falsely certifies z exactly when $\sigma \geq \sigma^*(z)$. Table S2 lists exact brackets on the diagonal $x = 1/2$; Figure 4(b) of the main text overlays them on the master-equation root.

Table S2: The critical elasticity $\sigma^*(z)$ on diagonal gap states $z = (1/2, s_1, s_2)$ (exact brackets; strict at both ends unless noted).

state $z = (x, s_1, s_2)$	$\sigma^*(z)$	note
(1/2, 1, 1)	∞ exactly	only the linear member certifies
(1/2, 3/2, 1/2)	∞ exactly	min coordinate ≤ 1: only-linear
(1/2, 6/5, 6/5)	$\in (2, 3)$, strict	the witness point
(1/2, 5/4, 5/4)	$= 2$ exactly	the $\theta = 1/2$ rung's equality state
(1/2, 13/10, 13/10)	$\in (1, 2)$	
(1/2, 3/2, 3/2)	$\in (1/2, 1)$, strict	the LPI-form witness
(1/2, 13/8, 13/8)	$\in (1/3, 1/2)$	the harmonic member's false-certification witness
(1/2, 9/5, 9/5)	$\in (1/5, 1/4)$	
(1/2, 39/20, 39/20)	$\in (1/17, 1/13)$	near the typed boundary: $\sigma^* \to 0$

Gap states with $\min(s_1, s_2) \leq 1$ are only-linear ($\sigma^* = \infty$); states with both coordinates > 1 have finite σ^*, given on the diagonal by the master-equation root, with $\sigma^*(s) \to 0$ as $s \to 2^-$ and $\sigma^*(s) \to \infty$ as $s \to 1^+$. Concretely, $\sigma = 3$ certifies $s = 6/5$, $\sigma = 2$ certifies $s = 13/10$, $\sigma = 1$ certifies $s = 3/2$, $\sigma = 1/2$ certifies $s = 13/8$ and $\sigma = 1/4$ certifies $s = 9/5$ (the per-rung false-certification witnesses of main-text Theorem 4(iii)); the harmonic member's Fibonacci pair is 8/5 rejects, 13/8 accepts.

S3.2. The exact off-diagonal criterion

On the viable gap quadrant $x < 1$, $1 < s_1, s_2 < 2$, the principal-plan trough inputs to the geometric member are positive. Write

$$A = \ln(s_1 - 1), \quad B = \ln(s_2 + 1), \quad C = \ln(s_1 + 1), \quad D = \ln(s_2 - 1).$$

The fast plan is licensed on a downward interval of weight share $p = w_1$, and the slow plan on an upward interval. Their union covers $[0, 1]$ exactly when the interval endpoints overlap.

Theorem S1 (off-diagonal geometric criterion)**.** *For $x < 1$ and $1 < s_1, s_2 < 2$, Protocol 2 at $\theta = 0$ accepts exactly when*

$$AD \leq BC, \qquad \textit{equivalently} \qquad r(s_1)r(s_2) \leq 1, \quad r(s) = \frac{\ln(1/(s-1))}{\ln(s+1)}.$$

Equivalently, its acceptance region is $s_2 \geq \rho(s_1)$, where $\rho = r^{-1} \circ (1/r)$ is a strictly decreasing involution. Its unique fixed point is $\sqrt{2}$, since $r(s) \leq 1$ iff $s^2 \geq 2$; its endpoint limits are $\rho(s) \to 2^-$ as $s \to 1^+$ and $\rho(s) \to 1^+$ as $s \to 2^-$. Thus $\min(s_1, s_2) \geq \sqrt{2}$ suffices for acceptance, while acceptance requires $\max(s_1, s_2) \geq \sqrt{2}$. For $\theta \in (0, 1)$, define $A = (s_1-1)^\theta$, $B = (s_2+1)^\theta$, $C = (s_1+1)^\theta$, and $D = (s_2-1)^\theta$; the analogous interval-cover condition is exactly

$$F := C(B-1) + A(1-D) - (B-D) \geq 0.$$

These are pointwise criteria; they do not assert a global fixed-sum phase structure.

Proof. At $\theta = 0$, the fast condition is $pA + (1-p)B \geq 0$, hence $p \leq U := B/(B-A)$; the slow condition is $pC + (1-p)D \geq 0$, hence $p \geq L := -D/(C-D)$. All denominators are positive on the stated quadrant. The cover is $L \leq U$, equivalent after cross-multiplication to $AD \leq BC$. Since $A, D < 0$ and $B, C > 0$, this is $r(s_1)r(s_2) \leq 1$. The function r is positive and strictly decreasing on $(1, 2)$, so ρ is well-defined and decreasing; the product

identity gives $\rho(\rho(s)) = s$. The fixed point is determined by $r(s) = 1$, which is equivalent to $s^2 = 2$, and the limits follow from $r(s) \to \infty$ at 1^+ and $r(s) \to 0$ at 2^-. The pivot statements follow because $r(s) \leq 1$ exactly for $s \geq \sqrt{2}$. For $0 < \theta < 1$, the same interval calculation gives $U = (B-1)/(B-A)$ and $L = (1-D)/(C-D)$; multiplying by the positive denominators and collecting terms yields $F \geq 0$. □

An exact harmonic specialization. At $\theta = -1$, the interval-cover expression simplifies to

$$F = \frac{2(s_1^2 - s_1 + s_2^2 - s_2 - 2)}{(s_1 - 1)(s_1 + 1)(s_2 - 1)(s_2 + 1)}.$$

The denominator is positive on $(1, 2)^2$, so acceptance is equivalent to $(2s_1 - 1)^2 + (2s_2 - 1)^2 \geq 10$. On the diagonal this reduces to $s \geq \varphi$, the harmonic critical floor. This identity follows algebraically.

S3.3. Scope of the off-diagonal results

The criteria above give pointwise acceptance conditions. They do not establish the minimum or stationary-point structure on a fixed-sum slice, equality of off-diagonal minima, monotonicity of the critical elasticity, or a global phase diagram. No such global claim is used in the main text. The diagonal thresholds are proved analytically in SI S2.

S3.4. The Living Planet Index identification, with its caveats

The $\theta = 0$ member has the geometric-mean functional form used by the Living Planet Index. Here it is applied to the constructed inputs $\lambda_i = 1 + s_i$, not to the published LPI abundance series. Each s_i is normalized by its own declared positive reference scale; consequently the critical thresholds depend on that normalization. Under the optional biomass map in main-text Section 5, $B_m/B_{\text{lim}} = 1 + s_1/2$, whereas the constructed index is $\lambda_1 = 1 + s_1$. They are not the same abundance ratio. The published LPI's input construction and fixed weighting are not Protocol 2; acceptance for every weight is more restrictive than acceptance at that one published weight, so the protocol's false-acceptance exposure can be smaller. No claim is made that the published LPI implements this assessment rule.

S3.5. Interpretation at the benchmark point

At the heatwave trough, the constructed values are $\lambda = (1/5, 11/5)$, giving an equal-weight linear mean $6/5$ and geometric mean $\sqrt{11/25} < 1$. This is a normalized-margin illustration of how the aggregator can alter a

report; it is not a claim about an operational LPI value or an automatic management response. The underlying biomass breach, terminal treatment, and income-margin assumptions are stated in SI S5.

S4. The common-shock variant in full

S4.1. The variant

Decompose the witness datum per the benchmark arithmetic: each principal plan's announced schedule is a mid-interval tent drawdown of depth $3/2$ in its characteristic coordinate (FAST: the quota pulse on s_1; SLOW: the gradual burden on s_2), the other coordinate constant; each environmental event is a transient tent dip of depth δ in one floor coordinate, troughing at $t = 1/2$ and recovered by $t = 1$, successors unchanged; the worst-case depth $2 = 3/2 + 1/2$ is the quota pulse plus the heatwave, so the datum's event depth is $\delta = 1/2$. STAGED spends x (it needs $x \geq 1$) while its floors grow linearly to $s + e$; an event of depth δ on a growing floor dips it to $s + 1/8 - \delta$ – by $\max(0, \delta - 1/8)$. The *common* class: the two coordinate events are separate disturbances, each striking every plan, so each plan's per-coordinate worst case takes its own schedule plus the event on that coordinate. The disturbance quantifier remains innermost. For STAGED, the aggregate dip $\max(0, \mathrm{raw}(p) - 1/8)$ is piecewise linear in the weight share p. Admissibility is therefore the conjunction of two affine inequalities on the branches; since the other plans' dips are affine in p, each cover condition reduces to endpoint evaluations.

Relation to the action-indexed case. At $\delta = 1/2$, the action-indexed classification and witness-point thresholds are those stated in main-text Theorem 2. The common- and coupled-shock variants below change the disturbance structure while retaining the declared plan menu and path semantics.

S4.2. The middle-cell verdict, the rescue trichotomy and blend fragility

These are main-text Proposition 6(i)–(iii), proved in S2.4. Two remarks complete them. The gap is strictly smaller than I by the exact $1/2$-cut: every action-indexed gap state with $\min(s_1, s_2) < 1/2$ is weak-rejected under the common class (witness $(\frac{1}{2}, \frac{1}{10}, \frac{19}{10})$, action-indexed weak-accepted by $s_1 + s_2 = 2$). And the blend window is class-conditional in general: under the action-indexed class at event depth δ it is nonempty exactly on $\{s_1 + s_2 \geq 3/2 + \delta\}$ (main-text Theorem 3 is the $\delta = 1/2$ instance), under the common class exactly on $\{s_1 \geq \delta,\ s_2 \geq \delta,\ s_1 + s_2 \geq 3/2 + 2\delta\}$; the coupled gap below, by contrast, is blend-closed.

S4.3. The coupled regime

Theorem S2 (the coupled regime, additive reading)**.** *Under the coupled class – one disturbance carrying both coordinate events simultaneously, riding on the plans' own schedules – at event depth δ: the typed accepted set coincides with the common class's at every depth, and*

$$\begin{aligned}\mathcal{V}^{\mathrm{cpl}}_{\mathrm{weak}} = \{x < 1,\ s_1 \geq \delta,\ s_2 \geq \delta,\ s_1 + s_2 \geq \tfrac{3}{2} + 2\delta\} \\ \cup \{x \geq 1,\ \min(s_1, s_2) \geq \max(0, \delta - \tfrac{1}{8})\}.\end{aligned}$$

At the datum's full magnitude ($\delta = 2$) every I-state is weak-rejected, but the claim is depth-dependent with exact boundary $\delta = 5/4$: at heatwave magnitude ($\delta = 1/2$) deep-I states survive as coupled gap states (witness $(\frac{1}{2}, \frac{19}{10}, \frac{19}{10})$); no I-state survives at $\delta \geq 5/4$, while $(\frac{1}{2}, \frac{79}{40}, \frac{79}{40})$ survives at $\delta = 5/4 - 1/40$; and at full magnitude the separation relocates rather than vanishes: the coupled gap $\{x < 1,\ 2 \leq s_1 < \frac{7}{2},\ 2 \leq s_2 < \frac{7}{2},\ s_1 + s_2 \geq \frac{11}{2}\}$ is nonempty (witness $(\frac{1}{2}, 3, 3)$ is weak-accepted and typed-rejected).

Proof sketch. Under the coupled class each plan's aggregate dip is the sum of both events' weighted contributions ($3p/2 + \delta$ for FAST) against the common class's maximum of single-event contributions. For $x < 1$, the endpoint weights $p = 0$ and $p = 1$ impose the separate coordinate cuts $s_2 \geq \delta$ and $s_1 \geq \delta$; the interior kink imposes $s_1 + s_2 \geq 3/2 + 2\delta$. The three conditions must be used together. For example, at $\delta = 1/4$, $(s_1, s_2) = (1/10, 39/20)$ passes the sum inequality but fails the $s_1 \geq \delta$ cut. An I-state has $s_1, s_2 < 2$, hence sum < 4, so no I-state survives when $\delta \geq 5/4$; the states $(79/40, 79/40)$ at $\delta = 5/4 - 1/40$ satisfy both coordinate cuts and the sum bound. The witness states realize the stated boundaries. □

Theorem S3 (the replace-reading collapse)**.** *Under the coupled-total class – the replace reading, each announced-schedule plan's worst case being the full-magnitude dip 2 in each floor, tent schedules suppressed – FAST and SLOW degenerate to the same weight-independent requirement $A(p) \geq 2$ (at the witness point both serve nowhere; at $(\frac{1}{2}, \frac{5}{2}, \frac{5}{2})$ both serve all of $[0, 1]$; at $(\frac{1}{2}, 3, 1)$ both serve exactly $[\frac{1}{2}, 1]$), and*

$$\mathcal{V}^{\mathrm{ct}}_{\mathrm{weak}} = \mathcal{V}^{\mathrm{ct}}_{\mathrm{typ}} = \{s_1 \geq 2,\ s_2 \geq 2\} \cup \{x \geq 1,\ \min(s_1, s_2) \geq \tfrac{15}{8}\} :$$

the typed/weak separation collapses entirely, and the replace reading is strictly more demanding than the action-indexed class (the witness $(\frac{1}{2}, 3, 1)$ is action-indexed typed-accepted yet a full reject here, as is the witness point itself).

Proof sketch. With the tents suppressed and the full dip 2 in each floor, every principal plan's aggregate dip is the constant 2, so the weak cover condition $A(p) \geq 2$ for all p is the pair of endpoint conditions $s_1 \geq 2$, $s_2 \geq 2$, which is the typed condition; STAGED's branch (the dip lands on the growing floor, threshold 15/8) is identical under both tests. □

The two readings are distinct stylized disturbance specifications; neither is established as an upper or lower bound on a physically realized coupled-shock process. They illustrate model sensitivity, not a bracket on empirical truth. Under the replace reading, the plan- and weight-independent dip removes the weak test's advantage of weight-wise plan choice, so the per-weight cover reduces to the per-coordinate condition used by the typed test.

Zero-disturbance case. At $\delta = 0$, the disturbance classes coincide in this stylized datum.

Model-relative consequences at the benchmark datum. These are conditional outputs of the constructed menu, disturbance classes, and path semantics, not empirical estimates or management prescriptions. The common-shock rule cuts the action-indexed weak gap at $\min(s) = 1/2$ (for example, $(\frac{1}{2}, \frac{1}{10}, \frac{19}{10})$); no reserve increment rescues failure states below the 3/8 floor cut. The coupled additive and replace readings yield distinct verdicts, neither claimed to bound a physical process. Under the common-shock rule, blended quota policies also fail on the specified band $s_1 + s_2 \in [2, \frac{5}{2})$. These are menu-relative feasibility results, not operational advice.

Verification and limits. The common- and coupled-class classifications follow from the analytic derivations above and in SI S2.4. Their scope is the specified menu, disturbance classes, and path semantics; they are not empirical estimates or bounds on realized shock processes.

S5. The resource-transition benchmark in full

Main-text Section 5 summarizes this synthetic illustration. Its biomass component is a Schaefer equation in model biomass units (MBU); the income-margin and reserve coordinates are normalized paths stipulated by the witness, not outputs of an economic model. The model is not fitted to a fishery. Published Northern-cod values are used only for the optional historical dimensional map below.

Table S3: Disturbance regimes on the witness datum and the status of the separation under each.

Regime	Disturbance class	Separation status
(i) Action-indexed separable	each plan's worst case is its own characteristic dip in the coordinate it moves	proven: main-text Theorem 2
(ii) Common shock	one shared shock strikes the same floor coordinate of every plan	computed (main-text Proposition 6): the gap survives, reduced by the exact 1/2-cut, licensing thresholds unshifted on the surviving region; the witness point remains a member
(iii) Coupled all-floor shocks	simultaneous dips across all floors of every plan	reading-conditional (Theorems S2–S3): additive – universal rejection exact on I at full magnitude, depth-dependent below $\delta = 5/4$, the gap relocating at full magnitude; replace – the typed/weak distinction itself collapses, with acceptance surviving on $\{s_1 \geq 2, s_2 \geq 2\} \cup \{x \geq 1, \min(s) \geq 15/8\}$

State, scales, and disturbance. Let $B_m(t)$ denote spawning biomass in MBU, with t measured in years and $\varsigma(B_m) = rB_m(1 - B_m/K)$, where $r = 4\,\mathrm{yr}^{-1}$, $K = 10$ MBU, and $B_{\mathrm{lim}} = 2$ MBU. The ecological margin is $s_1 = (B_m - B_{\mathrm{lim}})/(1\,\mathrm{MBU})$. The second coordinate is a normalized income margin $s_2 = \Delta\Pi/\Pi_0$, for a declared scale $\Pi_0 > 0$; no price, cost, or revenue equation is supplied. The reserve x is in normalized fund units, with staged cost $c = 1$. A heatwave strike, when present, subtracts $\delta_0 = 1/2$ MBU from biomass at $t = 1/2$; it does not affect s_2. The weighted dashboard acts on the dimensionless normalized coordinates, not on raw tonnes and currency.

The exact first-half schedule and the tube envelope. At $(x, s_1, s_2) = (1/2, 6/5, 6/5)$, $B_m(0) = 16/5$ MBU. On $0 \le t \le 1/2$, set $B_{\mathrm{line}}(t) = 16/5 - 3t$ and

$$H_m(t) = 3 + \varsigma(B_{\mathrm{line}}(t)), \qquad \dot{B}_m = \varsigma(B_m) - H_m(t).$$

The benign first half follows B_{line} exactly. The quota decreases from $1463/125 = 11.704$ to $2161/250 = 8.644$ MBU/yr, under the stipulated $H_{\max} = 13$ MBU/yr. At the midpoint the benign biomass is $17/10$ MBU; the heatwave branch drops to $6/5$ MBU. The certified floor paths are $s_1 : 6/5 \to -3/10 \to 6/5$ without the event and $s_1 : 6/5 \to -4/5 \to 6/5$ with it. The SLOW income-margin excursion is prescribed analogously, with biomass held at $16/5$ MBU. The STAGED path spends one fund unit and raises each normalized margin by $1/4$; its income and reserve trajectories are stipulated, not derived from cash-flow equations. The abstract action tubes and destination rules remain those listed in main-text Table 4.

With zero harvest after the midpoint, biomass follows the logistic equation rather than the piecewise-linear recovery envelope. From midpoint value b, the endpoint is

$$B_m(1) = \frac{10}{1 + (10/b - 1)e^{-2}}.$$

This gives $6.021359\ldots$ MBU from $b = 1.7$ and $5.018924\ldots$ MBU from $b = 1.2$, not the abstract successor $16/5$ MBU. The straight-line recovery is only a conservative lower envelope: until the realized stock reaches 5 MBU, $\varsigma(B_m) \ge \varsigma(6/5) = 528/125 > 4$ MBU/yr, exceeding the envelope's largest recovery rate; afterward the realized stock remains above 5 while the envelope is at most $16/5$. To use the abstract successor in a physical model, an additional end-of-interval control or reset must be specified. The comparison proves a biomass lower envelope, not the full two-coordinate transition or terminal attainability.

Income-margin dynamics and disturbance treatment. For the displayed witness, the illustrative schedules are: NO-SWITCH holds both normalized margins; FAST holds s_2 while s_1 follows its pulse-and-recovery tube; SLOW holds s_1 while s_2 follows the analogous depth-2 excursion; and STAGED increases both margins linearly by 1/4 while the fund is spent. The heatwave affects only s_1, and the declared no-event branch removes that biomass strike. These are complete specifications of the benchmark's second coordinate at the level of the witness tube, but not an empirically derived income model.

Dashboard verdict. On the heatwave branch of FAST, the normalized sum is $(12/5, 2/5, 12/5)$, so the equal-weight margin remains positive while the biomass floor is breached. At the canonical point, FAST is licensed for $r = w_2/w_1 \geq 2/3$ and SLOW for $r \leq 3/2$; the intervals overlap, and a biomass-only weight rejects FAST but can license SLOW. The claim is weight-adaptive state acceptance, not acceptance of one fixed trajectory under every weight. A biomass-limit breach is not, by itself, a claim of biological collapse. For states outside $\mathcal{V}_{\text{typ}}$ with $x < 1$, the reserve shortfall is $\kappa^* = 1 - x$ under the witness's staged action.

Table S4: The witness coordinates read as quantities in the illustrative benchmark.

Object	Illustrative reading
Typed margins $s_i \geq 0$	normalized biomass margin s_1 and separately normalized income-margin coordinate s_2
Scalarized operator E_w	dashboard $w \cdot (s_1, s_2)$ applied to declared dimensionless margins
Licensing thresholds ρ_1, ρ_2	weight-ratio ranges licensing the pulse (FAST) and gradual (SLOW) plans
Acceptance gap FP_{agg}	dashboard-certified yet typed-inadmissible transitions under the declared menu and disturbances
Rescue set R; threshold κ^*	states served by the staged plan and normalized fund increment needed to reach its cost
Blend versus concatenation	fractional physical control versus the separately stipulated union of complete primitive tubes
Exact-tube semantics	checking all states in the declared tube, including the midpoint event
CES elasticity σ^*	threshold in the constructed normalized-margin family, not a fitted biological parameter

Northern-cod anchor and dimensional map. DFO Science Advisory Report 2016/026, Table A2 (PDF p. 16), reports the assessment-vintage Northern-cod LRP as 884.6 kt, the rounded 1983–1989 mean spawning-stock biomass

of that series (884.58 kt from a total of 6192.06 kt). The 2015 value is 298.65 kt, about 33.8% of that historical LRP. This is not asserted to be a current-2026 threshold. If, only as a unit comparison, 2 MBU is mapped to 884.6 kt, then 1 MBU equals 442.3 kt:

Model quantity	MBU value	mapped physical value
Limit B_{lim}	2	884.6 kt
Opening biomass $B_m(0)$	16/5	1415.36 kt
Adverse trough	6/5	530.76 kt
Carrying capacity K	10	4423 kt
Heatwave strike δ_0	1/2	221.15 kt
Capacity $H_{\max}$	13 MBU/yr	5749.9 kt/yr

These large implied capacities make clear that the map is dimensional only; it does not make the model a plausible or fitted cod fishery. It is also important to preserve the index normalization: $B_m/B_{\text{lim}} = 1 + s_1/2$, while the CES construction uses $\lambda_1 = 1 + s_1$. If a physical biomass margin is $s_{1,\text{phys}} = 442.3 s_1$ kt, then $\lambda_1 = 1 + s_{1,\text{phys}}/442.3$, not $1 + s_{1,\text{phys}}$. The income coordinate uses its own scale, and the reserve has no physical currency unit here. No model parameter is estimated from DFO data, and the witness makes no empirical prediction.

Institutional interpretation of the reserve. The normalized threshold $\kappa^* = 1 - x$ can be read as an abstract adjustment-fund shortfall. Actual buy-backs, transition assistance, or quota banking require a monetary accounting model and an implementation mechanism not supplied by this illustration; these are examples of possible interpretations, not consequences of the theorem.

S6. Practitioner material

S6.1. Terminology map

S6.2. A ten-point audit checklist

For an agency using a composite dashboard to certify a sustainability transition, the four design tests of main-text Section 7.3 operationalise as ten audit points.

1. **Object of certification:** state what is being certified – a static state, an endpoint, or the existence of a path-wise safe plan.

2. **Binding floors:** list the floors that are legally binding, irreversible, or lack verified substitutability; these require noncompensatory (Protocol 4) treatment.

Table S5: Terminology map: assessment objects across three neighbouring vocabularies (1/2). The entries are correspondences of mathematical role, not analogies.

	MCDM / operations research	Control / viability theory	Environmental modelling / governance
Typed floors $s_i \geq 0$, path-wise	coordinate-wise criteria; per-criterion lower bounds along the path	state constraints; safe set; tube constraints	planetary boundaries; safe operating spaces; regulatory floors (for example, a biomass LRP)
Scalarized operator E_w	weighted-sum acceptance; weight space W_+	scalarized robust constraint on an aggregate output	composite sustainability index; the dashboard
Licensing thresholds ρ_1, ρ_2	weight intervals of acceptance; robustness of a recommendation to weights	parameter-dependent feasibility margins	indicator weighting schemes; stakeholder weight elicitation
Acceptance gap $\mathrm{FP}_{\mathrm{agg}}$	every weight licenses some plan, but no single plan meets all typed floors	weight-wise feasibility with no common admissible plan	aggregate acceptance under the stated rule despite typed rejection (menu-relative)

Table S6: Terminology map, continued (2/2).

	MCDM / operations research	Control / viability theory	Environmental modelling / governance
Rescue set R; threshold κ^*	alternatives repairable by a budgeted side payment	reachability under an auxiliary resource input	transition finance: adjustment funds or fleet buy-backs
Impossibility region I	accepted for every weighting, yet no common plan is feasible	empty common safe-action set; kernel exclusion	aggregate acceptance with a floor crossed mid-transition
Fractional blend vs. complete-excursion concatenation	physical fractional controls vs. the stipulated union of full primitive tubes	relaxed control vs. a declared switching class	blended schedule vs. complete primitive excursions
Exact-tube semantics	robust path-wise feasibility over a scenario set	viability tubes; worst-case reachability	avoiding transient breaches during a transition

3. **Policy quantifier:** declare whether the approval rule assumes one robust plan or lets the plan adapt to the weighting.

4. **Weight-dependent plans:** if weight-adaptive, report the plan associated with each relevant weight range (the thresholds ρ_1, ρ_2 are the pattern).

5. **Path-wise evaluation:** evaluate the floors along the full transition, not only at assessment snapshots.

6. **Disturbance structure:** specify whether shocks are common to all plans, action-indexed, or coupled across floors (Table S3); the separation's scope and the remedy ride on the class.

7. **Menu completeness:** acknowledge that negative feasibility results are menu-relative unless the action space is demonstrably exhaustive.

8. **Blended policies:** do not infer convexification from mathematical convenience; verify that fractional effort or quota allocation is administratively implementable.

9. **Rescue margins:** if a resource-controlled action exists, report the minimum increment κ^* that converts impossibility into rescue, and whether it is finite under the declared disturbance class.

10. **Mandatory reporting:** require that component trajectories and their minimum margins, the reserve and the acceptance protocol be reported alongside the aggregate, so that the typed verdict can be independently reconstructed; floors alone do not fix the side of the rescue threshold.

S6.3. A minimum report

A minimum report for a transition assessment should include: (1) typed floors and their provenance; (2) aggregate weights or the weight family; (3) policy quantifiers; (4) the disturbance set and class; (5) the observation and implementation model; (6) exact versus conservative tube status; (7) terminal maintainability; (8) action-exhaustion status; (9) uncertainty and approximation bounds; (10) normative premises.

S6.4. Data requirements for empirical application

Empirical application requires, in addition to the typed floors, disturbance set, plan menu, tube model and destination-maintainability witness, five further specifications.

1. **Action-set completeness status.** A negative certificate over a finite menu proves impossibility only relative to that menu. The application must state whether $\mathcal{A}$ is exhaustive, a listed policy menu, a sampled subset, or an inner approximation. If incomplete, the verdict is "no safe transition exists among the listed plans", not "no safe transition exists".

2. **Calibration and identifiability data.** Parameter estimates, uncertainty sets, structural alternatives, validation data, observation error, model discrepancy, missing-data treatment, disturbance dependence. A specified disturbance set is necessary but not sufficient; it must be justified, or the verdict marked conditional on model credibility.

3. **Policy and authority data.** Who may select each plan, what information they possess, decision timing, enforcement assumptions, compliance uncertainty, strategic responses, resource and legitimacy constraints.

4. **Threshold-uncertainty semantics.** A floor is rarely known exactly. Distinguish the deterministic floor $s_i \geq 0$ from robust threshold safety $s_i(z;\vartheta) \geq 0$ for all $\vartheta \in \Theta$, and from probabilistic or confidence-level versions if admitted.

5. **Destination maintainability status.** Whether the destination-maintainability witness is established physically, by simulation, or by assumption.

As a dimensional illustration only, the benchmark optionally maps 2 MBU to the historical 2016 Northern-cod LRP of 884.6 kt; this is not a parameter calibration or a claim about the current threshold.

S6.5. Endpoint evaluation on the witness

The typed-endpoint operator $E_{\mathrm{end,typ}}(z) = \{a : \forall d,\ \mathrm{End}(a,d) \subseteq S \text{ and } \mathrm{Succ}(a,d) \subseteq G\}$ satisfies $E_{\mathrm{typ}} \subseteq E_{\mathrm{end,typ}} \subseteq E_{\mathrm{end}}$. On the witness, FAST's endpoint values of s_1 equal the initial s_1 and its successor lies in G whenever $x \geq 0$, so $E_{\mathrm{end,typ}}(z) \neq \varnothing$ at every $z \in X_0$, while $E_{\mathrm{typ}}(z) \neq \varnothing$ holds exactly on $\{x \geq 1\} \cup \{s_1 \geq 2\} \cup \{s_2 \geq 2\}$: on the stricken slice $\mathcal{Q} \cap \{x < 1\}$ the endpoint evaluation is nonempty where the typed operator is empty – the photograph admits what the trajectory forbids. Reporting regimes that evaluate endpoints only (audited annual snapshots) therefore license transitions that violate typed floors mid-interval without detection; no empirical claim about any particular reporting regime is made.

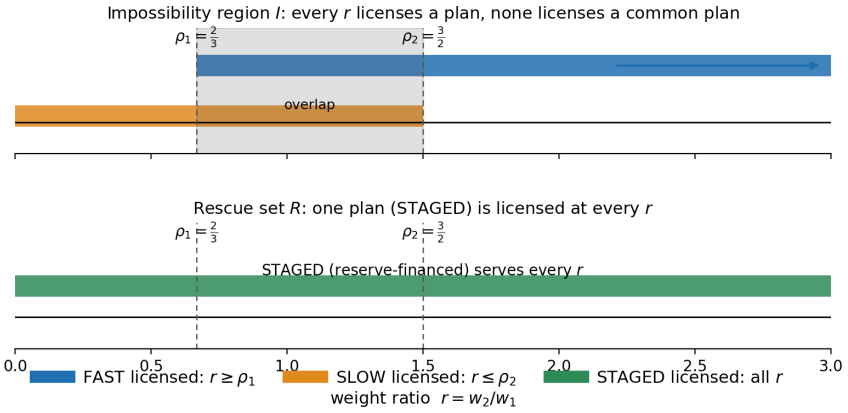


Figure S1: Weight-interval view at the witness point $(x, s_1, s_2) = (\frac{1}{2}, \frac{6}{5}, \frac{6}{5})$. FAST is licensed exactly for $r \geq \rho_1 = \frac{2}{3}$ and SLOW exactly for $r \leq \rho_2 = \frac{3}{2}$. The two intervals together cover every weight ratio $r \geq 0$, but neither covers it alone. On the rescue set, STAGED supplies the full interval and closes the gap.

S7. Numerical and formal verification scope

S7.1. Analytic results and finite checks

The analytic claims in the main text and SI S2–S4 are supported by the proofs supplied there. Finite calculations check specified cases; they do not establish continuum statements.

The exact-integer verifier cited in the main text enumerates 29,791 states (x, s_1, s_2) on a scale-40 grid and records 25 checks of witness classifications, region identities, licensing thresholds, endpoint comparisons, and hold-prefix persistence (Abaee, 2026d). This is exact arithmetic on a finite grid, not a proof over the continuum. The cited benchmark software reports 24 rational checks for the declared abstract action tubes and associated model quantities (Abaee, 2026f); it does not supply empirical income dynamics or calibrate a fishery model.

The diagonal CES thresholds and the pointwise off-diagonal geometric and harmonic criteria are proved in SI S2–S3. They do not establish a global fixed-sum stationary-point structure, equality of off-diagonal minima, monotonicity of the critical elasticity, or a global phase diagram. No such global claim is made.

S7.2. Formalization scope

A Lean source file for selected algebraic witness identities and CES rung examples is available at the cited immutable source revision: formalization

source. It does not formalize the general real-exponent nesting, continuity/IVT transfer, or relative-interior argument. Those steps are supported by the written mathematical proofs and are not claimed as machine-checked results.

S7.3. Data and code availability

No new empirical dataset was generated. The Northern-cod values are taken from DFO (2016), Table A2: the reported 1983–1989 mean is 884.58 kt, rounded to 884.6 kt, and the 2015 value is approximately 298.65 kt. They are used only as a historical dimensional comparison, not as a current threshold. The cited code records provide the finite checks described above but do not constitute a complete reproduction of every calculation reported here; all analytic claims are supported by the proofs in the article and Supplementary Information.

S8. Literature positioning

Commensurability. The foundation statement that values may be only weakly comparable – and that this incommensurability is constitutive of ecological economics – is due to Martinez-Alier, Munda and O'Neill (1998). The main article's operators give one formal model of the distinction that paper draws between strong and weak commensurability.

Maximin and indicator formulations. The characterization of sustainability indicators and the role of maximin and MSY-type reasoning in operationalizing strong-sustainability criteria are developed in Martinet (2011), Cairns and Martinet (2014), and Doyen and Gajardo (2020). This literature includes formal viability and Pareto formulations; the typed operator is one narrow transition-certification idealization and is not presented as a comprehensive or uniquely strongest formalization.

Compensability in multi-criteria decision analysis. Compensability analysis is established in MCDA, including mappings between compensatory aggregation, weak sustainability, and outranking methods (Cinelli, Coles and Kirwan, 2014; Schär, Pohl and Geldermann, 2025). Weighted-sum limitations on nonconvex Pareto fronts are also well known (Das and Dennis, 1997). The present finite-menu theorem shares convex-hull geometry with these results; it asks a different question about state acceptance and plan selection, and the same quantifier gap can occur for static alternatives.

Acceptance semantics and decision aiding. The protocols of main-text Table 1 are decision-aiding constructs: each specifies the rule by which a composite reading sorts a transition into the accepted or rejected category – the sorting problematic of decision aiding (Roy, 1996; Vincke, 1992). Under the weight-adaptive rule a transition is accepted when every weighting licenses some plan, possibly a different plan for each weighting; the licensing thresholds are then the verdict's weight-sensitivity intervals, analogous to sensitivity analyses applied to composite-indicator rankings (Nardo et al., 2008). The preference-elicitation background is multi-attribute value theory's substitution rates (Keeney and Raiffa, 1976) and the incommensurable-criteria treatment of sustainable development (Munda, 2005).

Viability and robust control. The backward recursion is a typed instance of robust-predecessor, reach–avoid, capture-basin and hybrid-reachability constructions (Aubin, 1991; Aubin, Bayen and Saint-Pierre, 2011; Saint-Pierre, 1994; Lygeros, Tomlin and Sastry, 1999; Aubin and Catté, 2002), with natural-resource management as an established application (De Lara and Doyen, 2008; Doyen et al., 2012). Fractional physical blending and complete-excursion concatenation compare distinct control classes; they are not a mixed/behavioral strategy equivalence. The here-and-now/wait-and-see analogy from adjustable robust optimization concerns the order of plan and weight quantifiers only; the weight is not modeled as a realized disturbance (Ben-Tal et al., 2004).

Planetary boundaries and the safe-and-just space. The typed operator is a formal idealisation of the separate-boundary reading underlying the Planetary Boundaries framework (Rockström et al., 2009; Steffen et al., 2015), and the same separately-checked-floor logic underlies the safe-and-just-space literature (Raworth, 2012; O'Neill et al., 2018; Fanning et al., 2022). Boundary processes interact and amplify human impacts (Lade et al., 2020), which is why the coupled disturbance regime of S4 is not a remote possibility. The main theorem provides a formal model of *one reason* a noncompensatory stance may be adopted; it does not prove that planetary boundaries ought to be noncompensatory, which is a normative and scientific-design question.

S9. Finite-generation consequence under compact confinement

A compactness consequence follows for nested generation constraints. Let K be a compact state set, and let $V^{(k)} \subseteq K$ be closed, nested decreasing generation constraints with $\bigcap_{k\geq 1} V^{(k)} = \varnothing$.

Proposition S1 (finite-generation obstruction)**.** *Under these assumptions there is a finite generation N for which $V^{(N)} = \varnothing$. Consequently, no trajectory confined to K can satisfy the requirements through every generation.*

Proof. If every $V^{(k)}$ were nonempty, the nested family of closed subsets of compact K would have the finite-intersection property. Compactness would imply $\bigcap_{k\geq 1} V^{(k)} \neq \varnothing$, contrary to the hypothesis. Hence some $V^{(N)}$ is empty. Since the sets are decreasing, every later generation set is also empty. □

The common compact confinement is essential to this finite-generation conclusion. Without it, take constant dynamics on $\mathbb{R}$ and $V^{(k)} = [k, \infty)$. Every finite prefix through generation N is feasible from $x = N$, but no state satisfies all generations. The initial state used for each finite prefix escapes every common compact set.

For fixed dynamics and a fixed policy class, if safe sets satisfy $K_1 \subseteq K_2$, then the viability kernel obeys $\mathrm{Viab}(K_1) \subseteq \mathrm{Viab}(K_2)$. Thus enlarging the safe set cannot contract the viability kernel under these assumptions. A different result would require a change in the dynamics, objective, or policy class.

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